

Alex Davison

Pittsburgh, PA, (919) 819-7153, alex@alexdavison.com, [LinkedIn](#), alexdavison.com

Software developer with a strong background in mathematics.

EDUCATION

Stony Brook University

- **Bachelor of Science in Mathematics**, Minor in Computer Science
- **Courses:** Computer Proofs, Algorithms, Cryptography (Graduate), Number Theory

Stony Brook, New York

Expected May 2027

SKILLS

- **Programming:** Python, TypeScript, C++
- **Tools:** Git, GitHub, Linux, Bash, numpy, matplotlib, $\text{\LaTeX}$
- **Math:** analysis, formal verification, algorithm design
- **Core Competencies:** project strategy, problem-solving, collaboration
- **Languages:** English (native), Mandarin (fluent), Russian (beginner)

EXPERIENCE

Gecko Robotics

Summer 2026

Software Intern

Solo-built the logbook pages and integration to Gecko's Cantilever software platform for power plants, encompassing all logging modes including daily operator rounds, offloading reports, and work orders. Integrated logbook context into Cantilever's agentic document search and LLM pipeline, and interviewed control room operators on-site at multiple plants, translating operator feedback into logbook features. Rolled out to 150+ plants in the US, comprising 5% of the North American power grid.

Reprogramming

Software Developer (Contracted)

2024

Designed and developed a JavaScript web application for Thales Academy, a network of 13 Pre-K-12 schools, automating creation of math worksheets for elementary students, with teacher-tailorable parameters to ensure alignment with curricula.

DataAnnotation

Freelance Software Developer

2023-25

Worked on a variety of script and codebase revisions across multiple languages (Python, Java, JavaScript, C, C++), providing thorough code reviews, troubleshooting approaches, and detailed explanations of revisions implemented.

Math Tutoring

Math Tutor

2023-now

Tutored students weekly on subjects ranging from elementary level math to calculus 4: private (2023-2024), Mathnasium (2024-2025), SBU ASTC (2025-present).

PROJECTS

ChessDB Mobile

2023

A mobile-friendly web app for exploring chess openings with live statistics and evaluation data from chessdb.cn and lichess.org. The original codebase behind the [ChessPath](#) program, with hundreds of users.

EOcross Solver

2022

A novel algorithm for Rubik's cube EOcross-solving that balances computer optimization with human intuition. The [CrystalCube](#) project implements my algorithm, used for training by several top speedsolvers.

STUDENT ENGAGEMENT

HackNYU 2025

2025

- Put together and led a four-person team. Guided brainstorming and project planning sessions.
- Developed front-end user profile UI, and back-end database support in Python with SQLite.
- Coordinated the merging of all contributors' work in Git/GitHub and produced the finalized project.

USA Computing Olympiad

2017

Competitive Chess – 1st place US Junior Chess Congress 2020, Stony Brook vs. Yale Invitationals 2024

2016-24